IMDB Movie Analysis

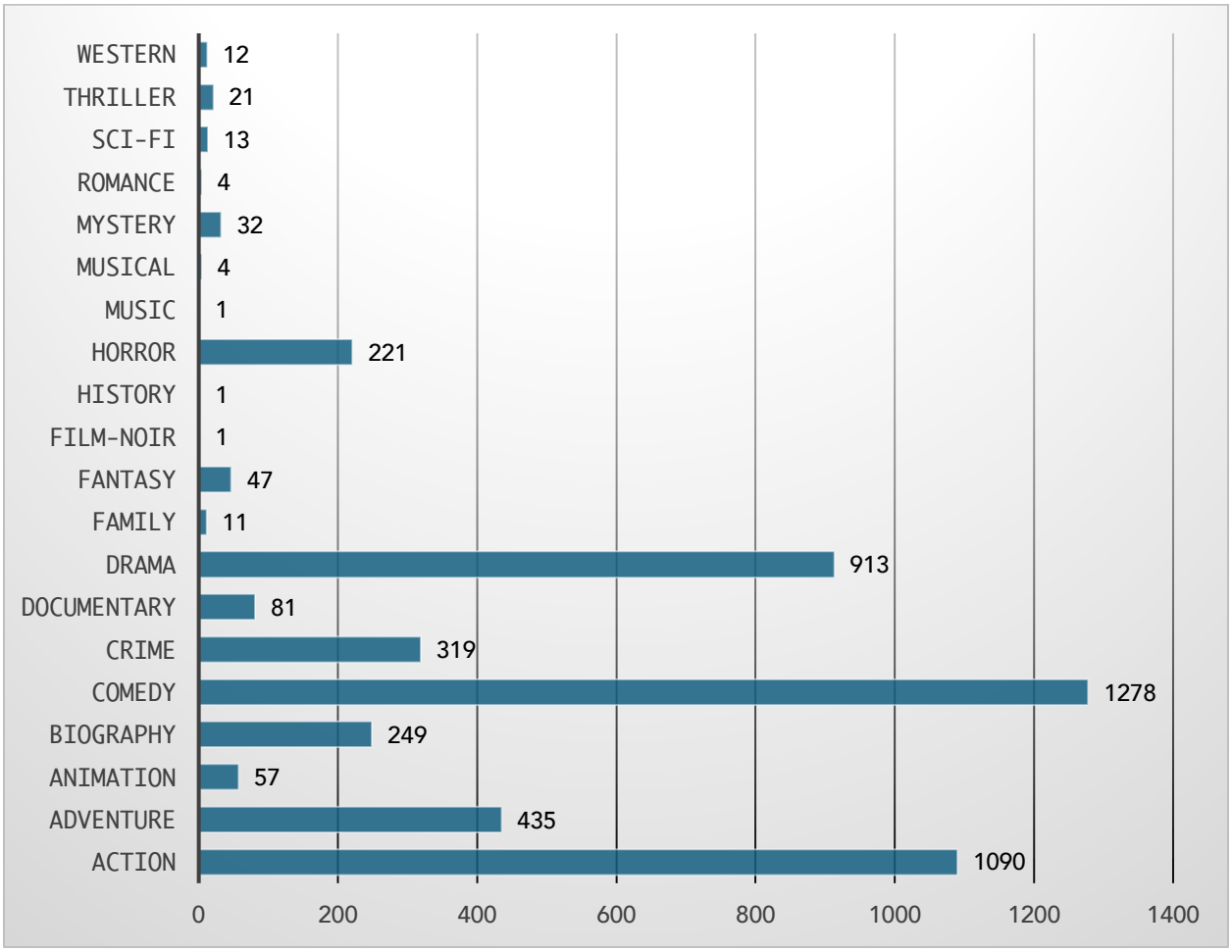
A. Movie Genre Analysis: Analyze the distribution of movie genres and their impact on the IMDB score.

- **Task:** Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.

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Genre	Count of IMDB Score
Action	1090
Adventure	435
Animation	57
Biography	249
Comedy	1278
Crime	319
Documentary	81
Drama	913
Family	11
Fantasy	47
Film-Noir	1
History	1
Horror	221
Music	1
Musical	4
Mystery	32
Romance	4
Sci-Fi	13

Thriller	21
Western	12



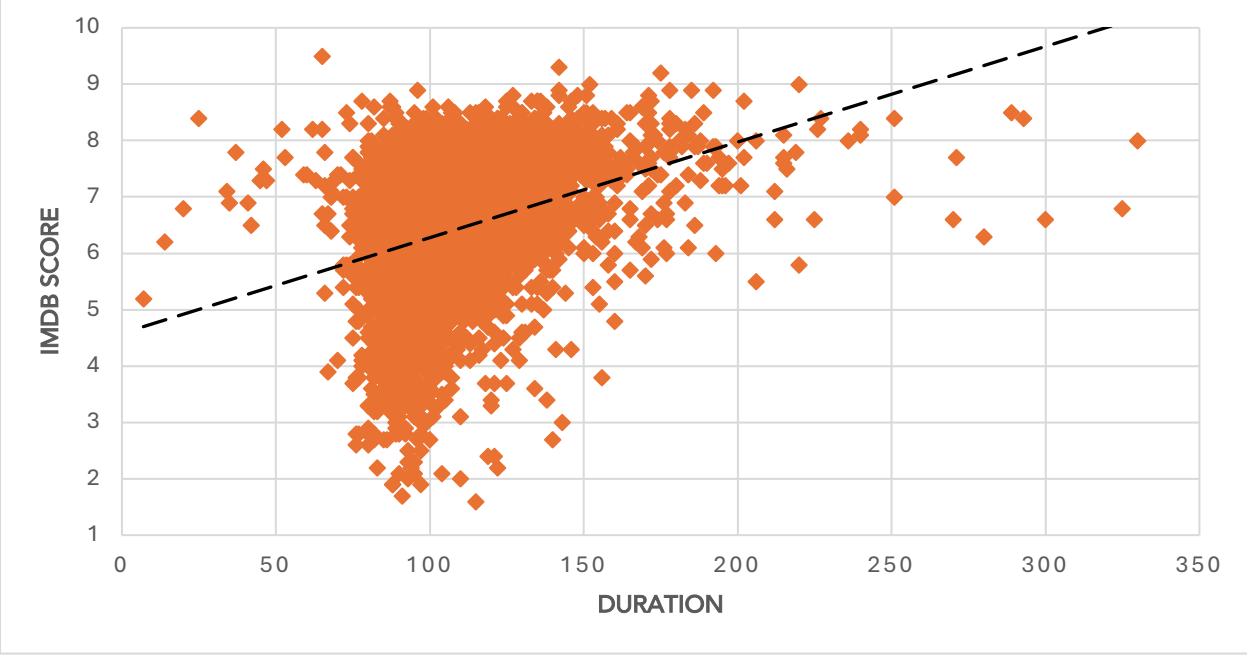
INSIGHT: Comedy genre has most movies followed by action & drama.

Genre	Count of IMDB Score	Average of IMDB Score	Var of IMDB Score	StdDevp of IMDB Score	Min of IMDB Score	Max of IMDB Score	Range of IMDB Score
Action	1090	6.2	1.2	1.1	1.7	9	7.3
Adventure	435	6.5	1.3	1.1	2.3	8.6	6.3
Animation	57	6.5	1.4	1.2	3.7	8.4	4.7
Biography	249	7.2	0.5	0.7	4.5	8.9	4.4
Comedy	1278	6.2	1.1	1.1	1.9	9.5	7.6
Crime	319	6.9	0.9	0.9	3.1	9.3	6.2
Documentary	81	7.2	1.4	1.2	1.6	8.7	7.1
Drama	913	6.7	1.0	1.0	2	8.8	6.8
Family	11	5.7	3.9	1.9	2.8	8.6	5.8
Fantasy	47	6.4	0.8	0.9	4.3	7.9	3.6
Film-Noir	1	7.6	#DIV/0!	0.0	7.6	7.6	0.0
History	1	7.5	#DIV/0!	0.0	7.5	7.5	0.0
Horror	221	5.6	1.2	1.1	2.2	8.5	6.3
Music	1	7.2	#DIV/0!	0.0	7.2	7.2	0.0
Musical	4	6.0	3.2	1.5	3.4	7.2	3.8
Mystery	32	6.5	1.3	1.1	3.2	8.5	5.3
Romance	4	6.3	0.6	0.7	5.2	7.1	1.9
Sci-Fi	13	6.0	2.2	1.4	2.8	8.2	5.4
Thriller	21	5.6	1.8	1.3	3.4	8.1	4.7
Western	12	6.6	2.6	1.5	3.8	8.9	5.1

B. Movie Duration Analysis: Analyze the distribution of movie durations and its impact on the IMDB score.

- Task: Analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score.

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Descriptive Statistics for Movie duration	
MEAN	108

MEDIAN	104
STANDARD DEVIATION	23

INSIGHT: Most movies are of 80-160 mins duration. As the duration increases, movies tend to score high IMBD ratings.

C. Language Analysis: Situation: Examine the distribution of movies based on their language.

- Task: Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

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Language	Count of Movie Name	Average of IMDB Score	StdDevp of IMDB Score
Aboriginal	2	7.0	0.6
Arabic	5	7.4	0.8
Aramaic	1	7.1	0.0
Bosnian	1	4.3	0.0
Cantonese	11	7.0	0.7
Chinese	3	5.7	0.4
Czech	1	7.4	0.0
Danish	5	7.5	1.0
Dari	2	7.5	0.1
Dutch	4	7.4	0.4
Dzongkha	1	7.5	0.0
English	4479	6.4	1.1
Filipino	1	6.7	0.0
French	72	7.0	0.7
German	19	7.3	0.9
Greek	1	7.3	0.0
Hebrew	5	7.6	0.3
Hindi	27	6.8	1.2
Hungarian	1	7.1	0.0
Icelandic	1	6.9	0.0
Indonesian	2	7.9	0.3
Italian	10	7.1	1.1
Japanese	16	7.4	1.0
Kazakh	1	6.0	0.0
Korean	7	7.4	0.8
Mandarin	24	6.8	1.0
Maya	1	7.8	0.0
Mongolian	1	7.3	0.0
Norwegian	4	7.2	0.5
Panjabi	1	6.6	0.0
Persian	4	7.6	1.0
Polish	1	7.4	0.0
Portuguese	8	7.5	0.8
Romanian	2	7.2	0.7
Russian	11	6.4	1.3
Slovenian	1	6.4	0.0
Spanish	40	6.9	0.8
Swahili	1	7.4	0.0
Swedish	5	7.4	0.7
Tamil	1	5.1	0.0
Telugu	1	8.4	0.0
Thai	3	6.6	0.4
Vietnamese	1	7.4	0.0
Zulu	2	7.1	0.2
Grand Total	**4790**	**6.4**	**1.1**

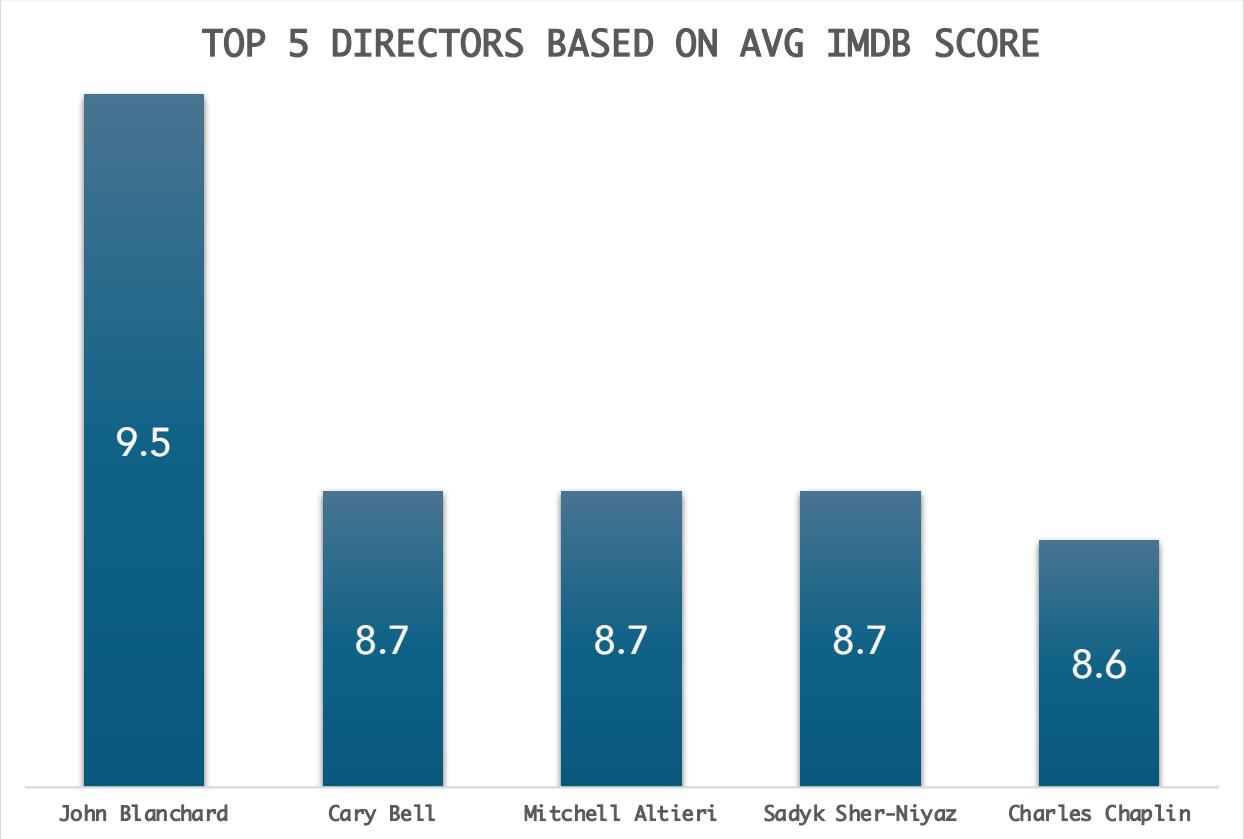
TOP 3 MOVIES		AVG	STD
English	4479	6.4	1.1
French	72	7.0	0.7
Spanish	40	6.9	0.8

INSIGHT: 94% of movies are produced in English language. Others are French & Spanish.

D. Director Analysis: Influence of directors on movie ratings.

- Task: Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

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INSIGHT: I used percentile function to filter directors who fall >= 99th percentile (8.3), and I found 29 directors whose IMDB score is 1.7% of the overall contribution.

E. Budget Analysis: Explore the relationship between movie budgets and their financial success.

- Task: Analyze the correlation between movie budgets and gross earnings and identify the movies with the highest profit margin.

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CORRELATION COEFFICIENT	0.24			Maximum Profit Margin	523505847
				AVATAR MOVIE GAINED MAX PROFIT	

INSIGHT: Correlation coefficient shows that, as budget of the movie increases, gross shows positive correlation because gross earnings too increases.

Profit Margin describes difference of budget from gross that states profit garnered by the movie where I found 'Avatar' movie has max. profit margin.